

Efficient jet-engine inlet RCS prediction using higher order FEM and reduced order modelling

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Abstract *A direct full wave based approach and a reduced order hp-FE sub-domain based scattering matrix methodology suitable for calculating the Radar Cross-Section (RCS) for electrically huge jet engine air intakes is presented. The convergence of the RCS using uniform mesh refinements and uniform polynomial enrichments are compared. The comparison shows a superiority in favour of the p-enrichments. An efficient method to avoid the bottleneck in the domain decomposition approach at the top-level is presented. A very efficient way is to solve the resulting interface problem using the presented reduced order modelling technique. The RCS can be computed with virtually no loss of accuracy in terms of generalised*

guided wave degrees of freedom. The auxiliary Generalised Scattering Matrix approach can also be used without the modal projection or with other generalised interfacial degrees of freedom to alleviate the interface bottleneck problem.

Keywords FEM, RCS, cavity, air intake, wave-guide, scattering matrix, equation solution, domain decomposition

1. Introduction

Reliable aircraft/target identification is a very important task especially in military applications. A non-responding target must be identified. The build-up of such a capability relies on the ability to predict the Radar Cross-Section (RCS) accurately. Measurements have revealed that the jet engine inlets are one of the major contributors to the overall RCS signature of an aircraft. Although the computational power is growing continuously the scattering characterisation of real aircraft configurations remain one of the most challenging problems in computational Electromagnetics (CEM). Jet engine inlets are often electrically huge semi-open channels. To hide the jet engine fan disks at the far end, the channel is often curved and partially coated with radar absorbing materials (RAM). The channel length may be hundreds of wavelengths, and at the same time its diameter may be tenths of wavelengths. Illuminating such a channel will create both guided

waves and standing waves. Due to the mixed attenuated and resonant behaviour, especially in the presence of dissipative coatings, the exact numerical method of choice is the finite element method (FEM). Since the volume is discretized with FEM it is possible to model and resolve the scattering from layers with material discontinuities and with the highly irregular jet engine fan blades. In the following we restrict our interest to the FEM which is exact in the sense that it converges to the solution of the continuous problem as the discretization is refined. Dealing with real configurations one must then be prepared for the solution of complex valued equation systems with effectively up to 100 million unknowns. Large efforts are therefore spent to make it feasible. It is understood that state of the art highly efficient parallelised computational sub-domain techniques have to be used to reliably predict the RCS of real jet engine air inlets. Although the RCS computations of such configurations are highly classified, there are published attempts to make this task tractable, see for example Barka et al. (2000). There are three main computational issues dealing with scattering on huge semi-open cavities,

- A) minimization of the number of unknowns of the basic discretized problem,
- B) efficient equation solution, and
- C) exploitation of possibilities to use reduced order modelling techniques.

Regarding A) we consider h -refinements where the element size is changed, and p -enrichments where the order of the element basis functions is changed. We use the most powerful FE-method, the so-called hp -version of FEM, allowing for both types of enhancements. It is well known that hp -FEM is able to maintain an exponential suppression of the discretization error measured in the natural norm of the problem versus the number of problem unknowns (degrees of freedom) even when the solution field is non-smooth. Of course, this requires an optimal highly problem dependent blend of h -refinements and p -enrichments. This can be achieved using automatic hp -adaptivity. These qualities are especially important in sections with material inhomogeneities and for modelling the scattering on the terminating section consisting of disks with irregular jet engine fan blades. In Zdunek and Rachowicz (2011) we investigated to some extent the possibility to benefit from the fast (asymptotically exponential) convergence that characterises the adaptive hp -FEM. For that purpose we used the $hp3d$ -code equipped with automatic adaptivity developed by L. Demkowicz and associates, Demkowicz (2007); Demkowicz et al. (2007). We predicted the cavity RCS using the novel and efficient half-space modelling technique described in Zdunek and Rachowicz (2008). We modelled the exterior problem mainly using $H(\text{curl})$ -conforming hp -version infinite elements (IE) comparing to an approach using newly developed

hp-version boundary elements (BEM).

The problem of removing hard bottlenecks connected with the equation solution, item B), is the the biggest, most difficult issue, and a very extensive topic by its own. We will deal with it in the form it appears in the present specialised case. In this investigation we use a state of the art open-source MPI and OPENMP parallelised so-called multi-frontal solver of the type described in Refs. Amestoy et al. (2001, 2006); Scott (2013). This type of single level domain-decomposition approach will face a memory and load balancing bottleneck at the top level if the fairly dense interface problem is solved in-core using one of the assigned MPI processes only. Here we solve systems with several million degrees of freedom using a two processor desk side PC with twelve cores. The mentioned bottlenecks can be alleviated using nested dissection domain decomposition and by solving the resulting top level dense problem separately, in parallel.

Regarding C), in Zdunek and Rachowicz (2011) we also provided cavity RCS predictions based on reduced order uncoupled interior problem analysis and the approximative Kirchhoff Aperture Integration (KAI) procedure. For homogeneous channel cross-sections we investigated the projection of the electric and magnetic fields on common section interfaces using a truncated TE- and TM-based wave-guide modal basis. We showed that substantial efficiency can be

gained in this way without significant loss of accuracy and reliability. The diffraction at the aperture rim is neglected in this case due to the KAI procedure. The comparison with the half-space model Zdunek and Rachowicz (2008) confirmed that the diffraction makes a difference but essentially only for illuminations at grazing incidence. We also showed that KAI approximations imply more severe restrictions regarding valid incidence angles.

In this contribution we summarise the findings in Zdunek and Rachowicz (2011) and add new results concerning item A). More precisely we compare RCS results using uniform h -extensions with uniform p -enrichments applied on a generic S-shaped perfectly conducting air-intake channel.

2. Model problem

Our model problem is described as an overall mathematical problem which is decomposed into several computational problems stated on non-overlapping subdomains. We start considering the overall mathematical problem. Let $\Omega \subset \mathbb{R}^3$ be a three dimensional channel like domain, for simplicity, describing a metallic (PEC) container whose interior mantle is possibly coated with a non-dissipative dielectric media equipped with a PEC lid at the far end. The front aperture is denoted Σ_0 and the back PEC lid is denoted Σ_m respectively.

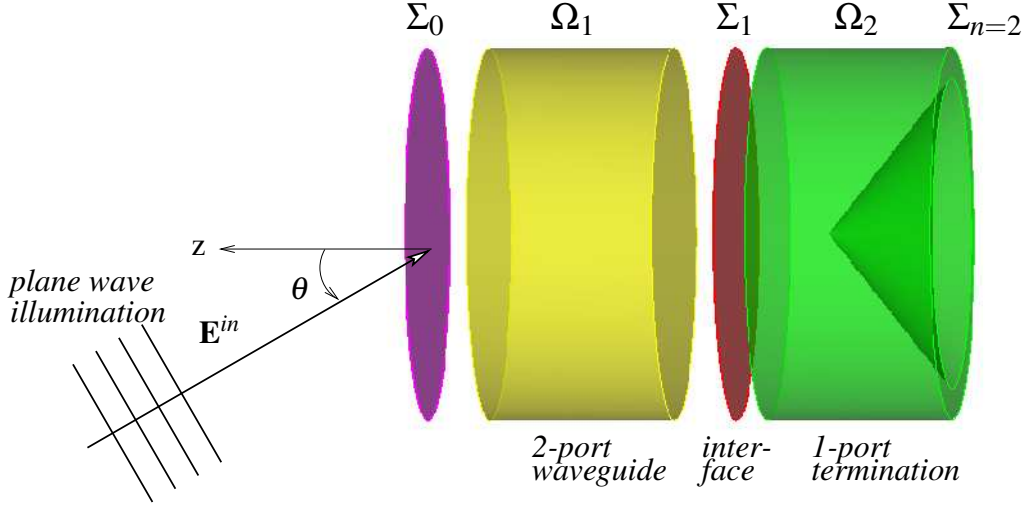


Figure 1: Semi-open channel with aperture Σ_0 and back lid Σ_2 . Decomposition into non-overlapping sub-domains Ω_1 and Ω_2 , with a common interface Σ_1 , such that $\Omega = \cup_{\ell=1}^m \Omega_\ell$ and $\Omega_1 \cap \Omega_2 = \emptyset$.

The real-valued relative dielectric permittivity $\epsilon_r = \epsilon_r(\mathbf{x})$ and magnetic permeability $\mu_r = \mu_r(\mathbf{x})$, are assumed to be scalar fields, respectively. The channel surface is assumed to be a sufficiently smooth surface $\partial\Omega$ admitting the definition of an exterior unit normal $\hat{\mathbf{n}} = \hat{\mathbf{n}}(\mathbf{x})$ at almost every point $\mathbf{x} \in \partial\Omega$. Let $\mathbf{M}_S$ be a given square integrable *tangential* magnetic surface current distribution on $\partial\Omega$

such that,

$$\mathbf{M}_S = \begin{cases} \mathbf{M}_{\Sigma_0}, & \text{on } \Sigma_0 \\ \mathbf{0}, & \text{on } \partial\Omega \setminus \Sigma_0, \end{cases} \quad (1)$$

In other words, $\mathbf{M}_S \in L_t^2(\partial\Omega) := \{\mathbf{u} \in L^2(\partial\Omega) : \mathbf{u} \cdot \hat{\mathbf{n}} = 0 \text{ on } \partial\Omega\}$.

On the aperture Σ_0 we assume the illumination to be given by a time-harmonic mono-chromatic plane linearly polarised electric field with amplitude E_0 ,

$$\mathbf{E}^{in} = E_0 \mathbf{p} \exp[i(\omega t - k_0 \mathbf{d} \cdot \mathbf{x})], \quad (2)$$

where $i = \sqrt{-1}$, the unit vector $\mathbf{p}$ defines the polarisation and $\mathbf{d}$ defines the direction of propagation, respectively. Further $k_0 = \omega/c_0$ is the free space wave number, wherein ω is the angular frequency and $c_0 = 1/\sqrt{\epsilon_0\mu_0}$ is the free space speed of light. Here ϵ_0 and μ_0 are the permeability and permittivity for free space respectively. The mathematical problem consists of solving the following boundary value problem in Ω :

Problem 1 (Strong time-harmonic magnetic formulation). *For given magnetic surface current $\mathbf{M}_S$ defined by Equation (1), find the magnetic field phasor*

$\mathbf{H} = \mathbf{H}(\mathbf{x}) \in \mathbb{C}$ that solves the pure Neumann problem:

$$\left\{ \begin{array}{ll} \nabla \times \varepsilon_r^{-1} \nabla \times \mathbf{H} - k_0^2 \mu_r \mathbf{H} = \mathbf{0}, & \text{in } \Omega, \\ \nabla \cdot (\mu_r \mathbf{H}) = 0, & \\ \hat{\mathbf{n}} \times \varepsilon_r^{-1} \nabla \times \mathbf{H} = -i\omega \varepsilon_0 \mathbf{M}_S, & \text{on } \partial\Omega. \end{array} \right. \quad (3)$$

By Ampere's law for a source free region and (3)₃ we have,

$$\mathbf{M}_S = \mathbf{M}_{\Sigma_0} = \mathbf{E}_t^{in} \times \hat{\mathbf{n}}, \quad \text{on } \Sigma_0. \quad (4)$$

□

Here and in the following a subscript $(_t)$ denotes the tangential trace $\mathbf{u}_t$, see Eq. (5), and where we we have used the identity $\mathbf{u}_t \times \hat{\mathbf{n}} = \mathbf{u} \times \hat{\mathbf{n}}$ in Eq. (4)₂.

Remark 1 (Tangential continuity). *In passing it is noted that given a surface Γ with oriented unit normal $\hat{\mathbf{n}}(\mathbf{x})$ we can uniquely split any electromagnetic field $\mathbf{u}$ defined on Γ into its tangential and normal traces as,*

$$\mathbf{u} \equiv \mathbf{u}_t + \mathbf{u}_n, \quad \mathbf{u}_t := (\mathbf{1} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}})\mathbf{u}, \quad \mathbf{u}_n := (\hat{\mathbf{n}} \otimes \hat{\mathbf{n}})\mathbf{u}, \quad (5)$$

wherein $\mathbf{1}$ is the identity mapping and $(\otimes)$ is the dyadic product. For $\mathbf{u} \in H(\mathbf{curl}, \Omega)$ only $\mathbf{u}_t$ need to be continuous across an interface Γ . This has implications for material discontinuities and for matching fields across computational domain interfaces, see Demkowicz (2007); Demkowicz et al. (2007).

The Gauss magnetic law Eq. (3)₂ asserts a solenoidal magnetic solution field $\mathbf{H}$ in the magneto-static limit as $k_0 \rightarrow 0$. Further, it should be noted that $\mathbf{H}$ is the so-called *total* magnetic field. Using that Maxwell's equations are linear we split the electromagnetic fields into incident and a back-scattered parts,

$$\mathbf{H} = \mathbf{H}^{in} + \mathbf{H}^{sc}, \quad \mathbf{E} = \mathbf{E}^{in} + \mathbf{E}^{sc}, \quad (6)$$

respectively. Given $\mathbf{E}^{in}$ we compute in a standard fashion the corresponding incident magnetic field in free-space using Faraday's law as,

$$\mathbf{H}^{in} = \frac{-1}{i\omega\mu_0} \nabla \times \mathbf{E}^{in}. \quad (7)$$

It can readily be seen that the incident field (7) satisfies the homogeneous reduced wave equation (3)₁ and the divergence condition (3)₂, in free-space, i.e. with $\varepsilon_r = \mu_r = 1$, respectively. Given $\mathbf{H}^{in}$ Problem 1 can either be solved as is for the total field or $\mathbf{H}$ for rewritten using Eq. (6)₁ and then solved for the back scattered field $\mathbf{H}^{sc}$ in such a fashion that the total field $\mathbf{H}^{in} + \mathbf{H}^{sc}$ satisfies the complete boundary value problem (3).

The corresponding variational formulations of Problem 1 formulated by defining the L^2 -inner product on Ω ,

$$(\mathbf{u}, \mathbf{v})_\Omega = \int_\Omega \mathbf{u} \cdot \bar{\mathbf{v}} \, dV, \quad \mathbf{u}, \mathbf{v} \in L^2(\Omega), \quad (8)$$

where the bar $(\bar{\cdot})$ denotes the complex conjugate operation, and also the inner product for traces $\mathbf{u}, \mathbf{v}$, defined on the boundary $\partial\Omega$ as,

$$(\mathbf{u}, \mathbf{v})_{\partial\Omega} = \int_{\partial\Omega} \mathbf{u} \cdot \bar{\mathbf{v}} \, dA \quad \mathbf{u}, \mathbf{v} \in L^2(\partial\Omega). \quad (9)$$

A Galerkin type variational formulation with improved stability in the magneto-static limit $k_0 \rightarrow 0$ is obtained following Demkowicz (2007) by including the divergence condition Eq. (3)₂. The result is:

Problem 2 (Stabilized variational magnetic formulation). Find $\mathbf{H}(\mathbf{x}) \in H(\mathbf{curl}, \Omega)$

and Lagrange multiplier $p \in W$:

$$\begin{cases} b(\mathbf{H}, \mathbf{F}) + d(p, \mathbf{F}) = l(\mathbf{F}), & \forall \mathbf{F} \in \mathbf{Q}, \\ d^*(\mathbf{H}, q) = 0, & \forall q \in W, \end{cases} \quad (10)$$

where $d^*(\mathbf{H}, q) = \overline{d(q, \mathbf{H})}$, where b , d and l are the following continuous sesquilinear and antilinear forms, respectively,

$$b(\mathbf{H}, \mathbf{F}) := (\boldsymbol{\varepsilon}_r^{-1} \nabla \times \mathbf{H}, \nabla \times \mathbf{F})_\Omega - k_0^2 (\boldsymbol{\mu}_r \mathbf{H}, \mathbf{F})_\Omega \quad (11)$$

$$d(p, \mathbf{F}) := -k_0^2 (\boldsymbol{\mu}_r \nabla p, \mathbf{F})_\Omega, \quad (12)$$

$$l(\mathbf{F}) := -i\omega\boldsymbol{\varepsilon}_0(\mathbf{E}^{in} \times \hat{\mathbf{n}}, \mathbf{F})_{\Sigma_0}, \quad (13)$$

and where,

$$\mathbf{Q} := \{\mathbf{F} \in H(\mathbf{curl}, \Omega) : \hat{\mathbf{n}} \times \mathbf{F} = \mathbf{0} \text{ on } \partial\Omega_D\},$$

and

$$W := \{\phi \in H^1(\Omega) : \phi = 0 \text{ on } \partial\Omega_D\},$$

wherein $\partial\Omega_D$ denotes the part of the boundary where Dirichlet type conditions are given. $\square$

Here we have a pure Neumann $\mathbf{H}$ -field problem with $\partial\Omega_D = \emptyset$. The stabilized formulation is possible because the gradient of a scalar field in $H^1(\Omega)$ form precisely the null space of the curl-curl operator. In passing it is recalled that $\mathbf{H}$ is the *total* magnetic field.

Full wave response analysis. In direct full wave analyses we use an equivalent $\mathbf{E}^{sc}$ based scattered field approach and consider the channel with its aperture Σ_0 in a PEC ground plane $z = 0$. The exterior is then represented with a truncated half-spherical mesh coated with infinite elements that model the far-field radiation. See Zdunek and Rachowicz (2008) for details.

Electrically large channels. For any computationally large problem the domain Ω is decomposed into sub-domains $\Omega_\ell, \ell = 1, 2, \dots, m$ with common interfaces $\Sigma_\ell, \ell = 1, 2, \dots, m-1$, essentially as depicted in Figure 1. The number of domains and their sizes are dictated by the geometry, the distribution of RAM coatings, the number of unknowns that can be handled by the actual computer. That is,

its memory size, the number of processors, the number of threads per processor, available disk-space etc. In our case we use a multi-frontal solver of the type Scott (2013). It is set to eliminate all interior degrees of freedom in each of the sub-domains, leaving the traces on the common interfaces and on the aperture to the top-level interface problem. If the overall interface problem fits into core we can solve it directly to obtain the sought back-scattered magnetic field $\mathbf{H}^{sc}$ on the aperture Σ_0 . If it is a full wave direct response analysis one of the domains will be the exterior domain. If we use our reduced order approach we do not model the exterior outside the channel and the interface problem is solved in a separate solution step, as explained below. In any case, having the electromagnetic currents on the channel aperture computations to get the RCS are standard, see Demkowicz (2007); Demkowicz et al. (2007). There are a number of ways to avoid assembling and solving the overall interface problem in core. One is obviously to replace the interface problem by a new level of elimination of the unknowns at a common interfaces starting from the end of the channel and proceeding to the aperture. This is a very natural solution for a frontal type of solver. Here we present such an approach together with our reduced order modelling technique.

Generalised scattering matrix based reduced order modelling. In our reduced order modelling approach Zdunek and Rachowicz (2011) we use the simple Kirchhoff aperture integration approximation. For clarity, in this paragraph, we use the coordinate system and the model configuration depicted in Figure 1. Scattering exterior to the channel outside the datum $z = 0$, see Figure 1, is not meshed and solved for. Our reduced order modelling approach substantially reduces the size of the interface problem. It provides a very promising way to avoid the hard computational bottleneck with a too large overall interface problem. The size reduction is achieved using the concept of travelling and attenuated waves in the channel. We consider a representation of the full three-dimensional electromagnetic fields using the following ansatz with two-dimensional phasor fields,

$$\begin{pmatrix} \mathbf{E}(x, y, z) \\ \mathbf{H}(x, y, z) \end{pmatrix} = \begin{pmatrix} \mathbf{E}(x, y) \\ \mathbf{H}(x, y) \end{pmatrix} \exp[\pm i k_z z], \quad (14)$$

for z fixed, where k_z is the axial propagation constant to be determined, which may be real or complex. Guided and standing waves in the z -direction can be formed by appropriate linear combinations. This ansatz may not be as good exactly at the aperture $\Sigma_0(z = z_0)$. Using ansatz (14) it is advantageous to separate the two-dimensional phasor fields $\mathbf{E}(x, y)$ and $\mathbf{H}(x, y)$ into tangent traces $(\mathbf{E}_t(x, y), \mathbf{H}_t(x, y))$ and normal traces $(\mathbf{E}_z(x, y), \mathbf{H}_z(x, y))$ on the actual interface

using (5). The size reduction is achieved by replacing the finite element degrees of freedom on the aperture and on the interfaces between the computational domains, see Figure 1, by generalised degrees of freedom that represent the participation factors (a_n, b_n) of in-coming and out-going guided waves. For z fixed we have,

$$\mathbf{E}_t(x, y, z) = \sum_{n=1}^N \mathbf{E}_{t,n}(x, y) (a_n \exp[-ik_{z,n}z] + b_n \exp[+ik_{z,n}z]). \quad (15)$$

Here $(\mathbf{E}_{t,n}, k_{z,n})_{n=1}^N$ are the N propagating eigen-fields and associated purely imaginary propagation constants $k_{z,n} = i\beta_{z,n}$ determined at the actual interface. Propagation constants $k_{z,n}$ for $n > N$ correspond to heavily damped so-called evanescent modes. The dimensions of the cross-section determine the so-called cut-off frequency ω_c above which waves cannot propagate. The finite number of propagating modes N provides a natural truncation for series (15). Lightly damped configurations are described taking cN , with $c > 1$. Thus, this approach implies solving the homogeneous problem corresponding to (10) at m fixed datums z , Σ_ℓ , $\ell = 0, m-1$. This approach allows treating materially inhomogeneous channel cross sections Σ , see Vardapetyan and Demkowicz (2002); Vardapetyan et al. (2003) and confer with Lee et al. (1991). For materially homogeneous sections the solution for the eigen-functions and propagation constants simplifies to the standard determination of the TE- and TM-modes and their propagation constants,

respectively. In any case, the approach relies on the linear magnetic Neumann-to-Dirichlet operator (NtD) $\mathcal{G}_{\mathbf{H}}[\bullet]$ related to Problem 1:

$$\mathcal{G}_{\mathbf{H}}[\mathbf{E}_t \times \hat{\mathbf{z}}] = (\mathbf{H}_t \times \hat{\mathbf{z}})|_{\Sigma_\ell}, \quad \text{on } \Sigma_\ell, \quad \text{for } \ell = 0, 1, \dots, m-1 \quad (16)$$

where $\mathbf{H}_t$ are the tangential traces of the solution of Eqs. (10) at Σ_ℓ , $\ell = 0, 1, \dots, m-1$.

It is readily seen that $\mathcal{G}_{\mathbf{H}}$ corresponds to a Generalised Admittance Matrix (GAM) operator with units $[A/V]$. An L^2 formulation including the NtD-operator (16) is obtained by projecting on the electric tangential trace $\mathbf{E}_t$. The result is,

$$(\mathbf{E}_t, \mathcal{G}_{\mathbf{H}}[\mathbf{E}_t \times \hat{\mathbf{z}}])_{\Sigma_\ell} = (\mathbf{E}_t, \mathbf{H}_t \times \hat{\mathbf{z}})_{\Sigma_\ell}, \quad \text{for } \ell = 0, 1, \dots, m-1. \quad (17)$$

Having $(\mathbf{E}_t, E_z)$ we can compute $(\mathbf{H}_t, H_z)$ using the relation,

$$\begin{cases} \mathbf{H}_t &= \frac{k}{\omega\mu_r} \hat{\mathbf{z}} \times \left(\mathbf{E}_t + \frac{1}{ik_z} \nabla_t E_z \right) \\ H_z &= \frac{i}{\omega\mu_r} \hat{\mathbf{z}} \cdot (\nabla_t \times \mathbf{E}_t) \end{cases} \quad (18)$$

A similar relation is readily obtained for the reverse situation. Relation (18) is given for waves propagating in the positive z -direction. The truncated series ex-

pansion in eigen-functions in Eq.(15) is rewritten in vector form as,

$$\mathbf{E}_t(x, y, z) = \mathbb{E}_t \mathbf{a} + \mathbb{E}_t \mathbf{b}, \quad (19a)$$

$$\mathbb{E}_t = [\mathbf{E}_{t,1}, \mathbf{E}_{t,2}, \dots, \mathbf{E}_{t,N}], \quad (19b)$$

$$\mathbf{a} = [a_1 \exp(-ik_{z,1}z), a_1 \exp(-ik_{z,2}z), \dots, a_1 \exp(-ik_{z,N}z)]^T, \quad (19c)$$

$$\mathbf{b} = [a_1 \exp(+ik_{z,1}z), a_1 \exp(+ik_{z,2}z), \dots, a_1 \exp(+ik_{z,N}z)]^T. \quad (19d)$$

Using (19) can now be used to recast Eqs. (17) into the following algebraic form,

$$\mathbb{K}_\ell(\mathbf{a} + \mathbf{b}) = \mathbb{D}_\ell^{f(+ik_z)} \mathbf{a} + \mathbb{D}_\ell^{f(-ik_z)} \mathbf{b}, \quad (20a)$$

$$\mathbb{K}_\ell = (\mathbb{E}, \mathcal{G}_\mathbf{H}[\mathbb{E}_t \cdot \times \hat{\mathbf{z}}])_{\Sigma_\ell}, \quad (20b)$$

$$\mathbb{D}_\ell^{f(u_n)} = f(u_n) \delta_{nm}, \quad n, m = 1, 2, \dots, N, \quad (20c)$$

where $f(u_n)$ is obtained using (18) with the actual propagation constant and wave direction, δ_{nm} is the $N_\ell \times N_\ell$ Kronecker symbol, wherein N_ℓ is the number of propagating waves through interface Σ_ℓ , and where

$$\mathbb{E}_t \cdot \times \hat{\mathbf{z}} := [\mathbf{E}_{t,1} \times \hat{\mathbf{z}}, \mathbf{E}_{t,2} \times \hat{\mathbf{z}}, \dots, \mathbf{E}_{t,N} \times \hat{\mathbf{z}}]. \quad (21)$$

Finally, the participation factors for the out-going guided waves $\mathbf{b}$ are computed by rewriting and solving Eq. (20a) for a given incident field $\mathbf{E}^{in}$ (2) represented by

the guided waves with participation factors $\mathbf{a}$ at the aperture Σ_0 ,

$$\mathbb{S}_\ell \mathbf{b} = \mathbf{a}, \quad (22a)$$

$$\mathbb{S}_\ell = -(\mathbb{K}_\ell - \mathbb{D}_\ell^{f(-ik_z)})^{-1}(\mathbb{K}_\ell - \mathbb{D}_\ell^{f(+ik_z)}). \quad (22b)$$

Here $\mathbb{S}_\ell$ is the so-called $(N_\ell \times N_\ell)$ Generalised Scattering Matrix (GSM) for a single interface Σ_ℓ . The GSM approach is readily applied to Sections Ω_ℓ , $\ell = 1, m-1$ with two ports $\Sigma_{\ell-1}$ and Σ_ℓ . The reduced order model interface problem can be solved very efficiently by solving the overall $\mathbb{S}$ -matrix formulation in a frontal fashion for the participation factors of the out-going guided waves at the aperture Σ_0 .

3. Numerical examples.

First we investigate the RCS convergence using low order h -type mesh refinements compared to using uniform enrichments of polynomial degree, for an S-shaped metallic cavity. This investigation concerns computing RCS with the minimal number of unknowns of the basic discretized problem (item A) in Section 1). In this part we use the full wave approach modelling both the interior channel scattering and the scattering exterior to the channel as described in Section 2. The resolution of the model geometry and the resolution of the electromagnetic fields

imposes requirements on the mesh density and/or the polynomial degree of the finite elements used which are taken into consideration.

Next we report on the possibility to use a GSM wave-guide modal reduced order modelling technique for efficient RCS computation without loss of accuracy. For that purpose we use a simple straight cylindrical PEC cavity with a conical termination. The approach uses a domain-decomposition technique which enables an efficient equation solution in parallel (item B) in Section 1). The resulting so-called interface problem is here solved in an auxiliary step removing a possibly hard bottleneck of standard one-level multi-frontal solvers. Moreover, it greatly saves on the number of degrees of freedom (item C) in Section 1) using the guided wave GSM approach described in Section 2.

S-shaped PEC cavity. The S-shaped PEC cavity used for the RCS convergence tests is described in Figure 2. A nominal $p=2$ mesh, with quadratic edge based brick elements that resolves the channel geometry fairly is used as starting point. The nominal $p=2$ mesh does not resolve of the electromagnetic fields as good, at the frequency $f = 6$ GHz, as it resolves the geometry. The resolution of the channel geometry is thus fixed by the density and interpolation order of the nominal mesh.

Figures 3 and 5 show the outcome in terms of the RCS raising the polynomial order uniformly from $p=4$ to $p=5$ for horizontal and vertical polarisation, respectively. Note that these two enriched models have 1.2 and 2.3 million complex valued degrees of freedom (M dof), respectively. The (a) parts of the Figures show mono-static RCS versus horizontal azimuth angle θ , as defined in Figure 2. Parts (b) of these Figures show the RCS difference between the $p=4$ and $p=5$ solutions. They provide an estimate of the RCS-error. It is seen that the 1.2 M dof $p=4$ solutions are close the the $p=5$ solutions in the interval $\theta \in [-90, +60]$ for both polarisations. However, the RCS-error varies substantially with the angle θ . It is also noted that the accuracy of the $p=4$ model is not equal for horizontal and vertical polarisation.

Figures 4 and 6 show the outcome in terms of the RCS, breaking each nominal brick element into eight (8) bricks uniformly. The operation of one such uniform mesh refinement is denoted $1 \times h8$. It is seen that the RCS computed with the nominal $p = 2$ mesh with a single $1 \times h8$ is not converged compared to the RCS computed with that with a uniform $p = 4$ extension. As above the (a) parts of the Figures show mono-static RCS versus horizontal azimuth angle θ and the (b) parts show the RCS difference relative to the uniform $p = 4$ result. It is noted that the $p = 2$ mesh with a single $1 \times h8$ refinement has almost the same number

of unknowns as the uniform $p = 4$ extension, i.e. 1.2 Mdof. In Figure 4 we also show the RCS result obtained raising the polynomial order uniformly to $p = 3$ in the nominal mesh and then applying a single $1 \times h8$ refinement. The resulting model has almost 4 Mdof. The RCS result is very close to that obtained with the 1.2 Mdof uniform $p = 4$ model.

As expected from theory for smooth solutions, we find that uniform p -extensions are much more efficient measured with respect to the number of unknowns (dof:s) comparing with uniform h -refinements. It should be noted that p -extensions reduces the sparsity of the equation system, which using Gaussian elimination, enlarges the solution time and the memory requirement. These adverse effects should be taken into consideration considering the overall equation solution efficiency. In our case the wall clock time for solving the uniform $p = 3$ plus $1 \times h8$ refinement was markedly longer compared to the wall clock-time for solving the uniform $p = 4$ model.

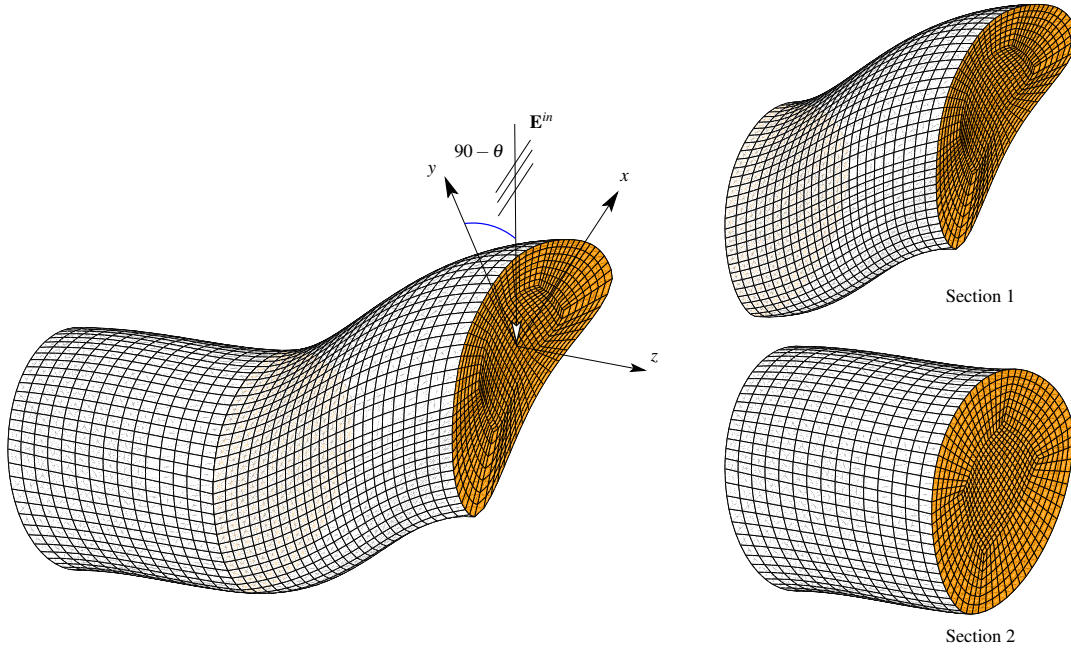
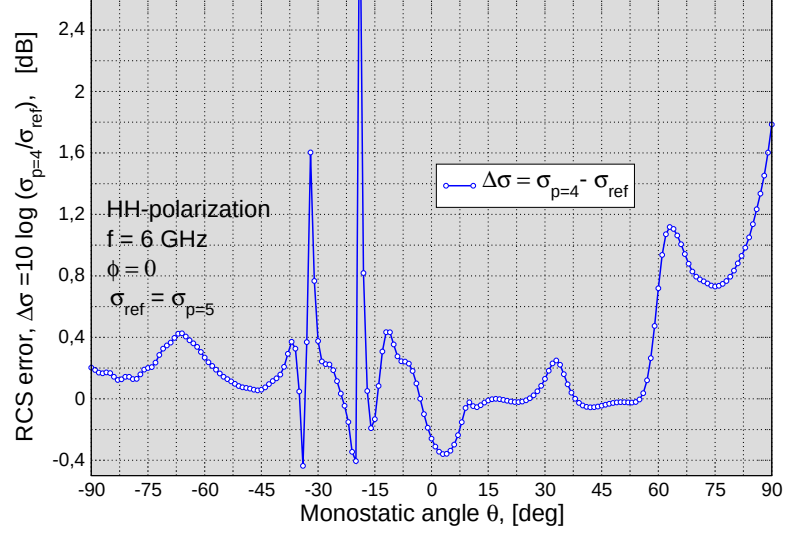
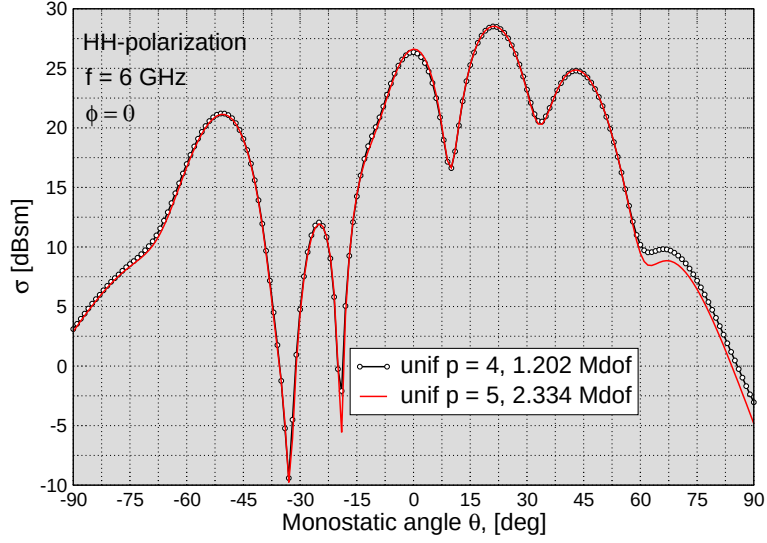


Figure 2: S-shaped PEC cavity $L = 1.228 [m]$. PEC Engine Aperture $D = 0.264 [m]$. Inlet aperture $w = 0.38 [m]$, $h = 0.18 [m]$. Plane wave illumination. About 210 propagating TE-modes and 180 propagating TM-modes @ 9 GHz.



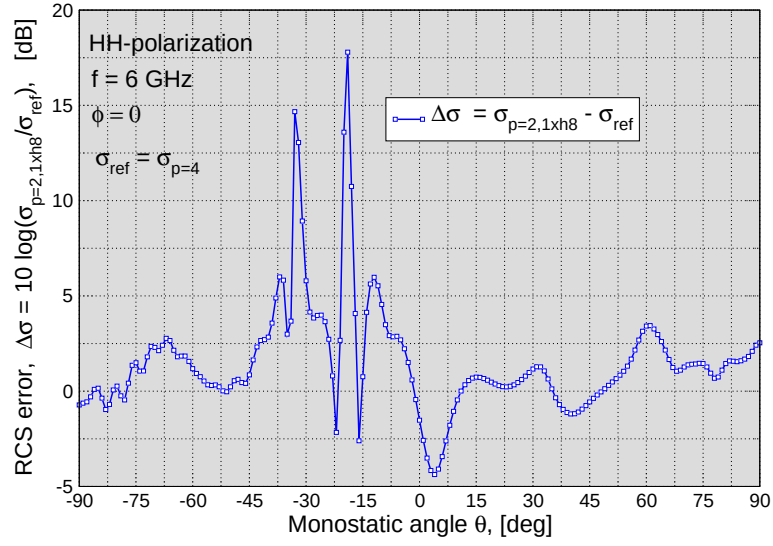
(b)



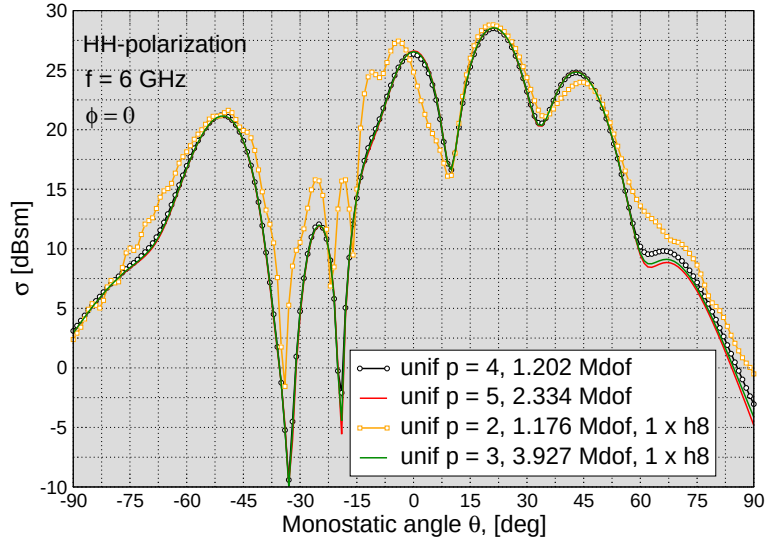
(a)

Figure 3: S-shaped PEC cavity. Mono-static RCS versus azimuth angle θ at 6 GHz. Horizontal polarisation. RCS convergence using uniform p -enrichments.

(a) Uniform $p=4$ and $p=5$. (b) RCS error estimate

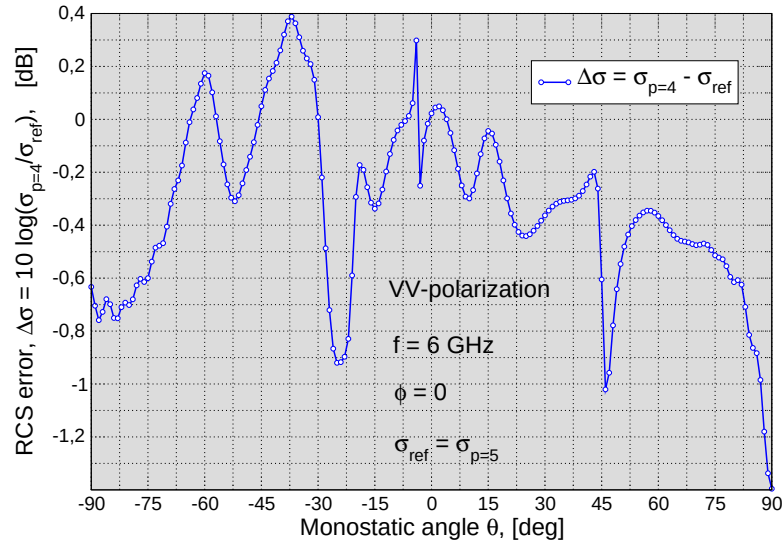


(b)

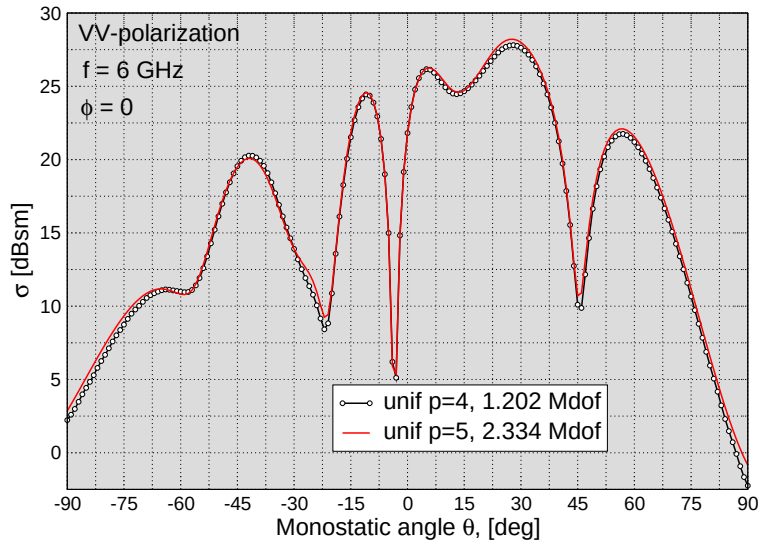


(a)

Figure 4: S-shaped PEC cavity. Mono-static RCS versus azimuth angle θ at 6 GHz. Horizontal polarisation. RCS convergence using uniform h -extensions. (a) Uniform $p=4$ and a uniform $p=2$ mesh with a $1 \times h8$ refinement. (b) RCS error estimate

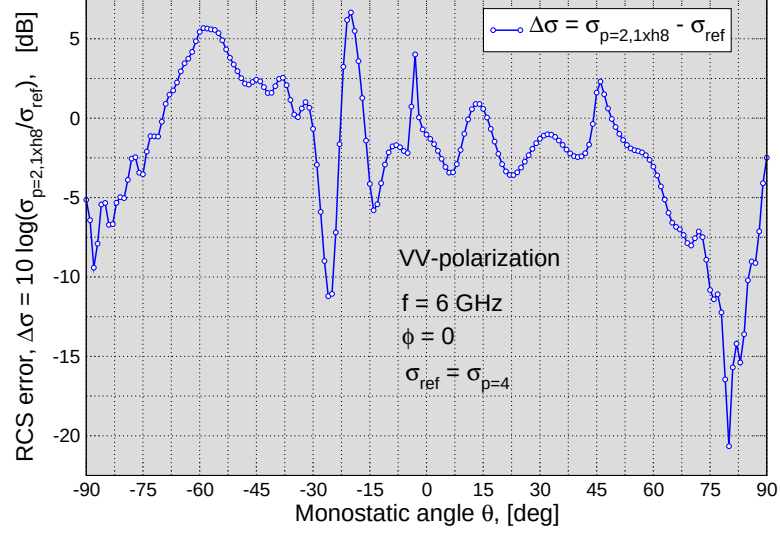


(b)

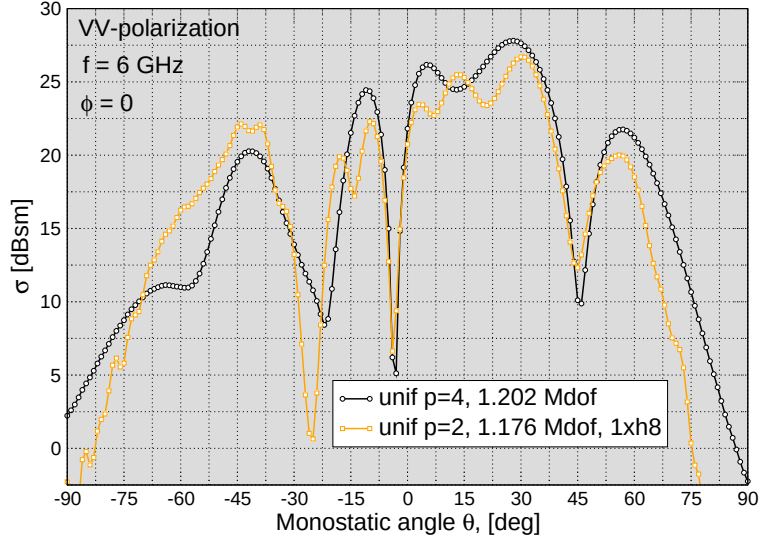


(a)

Figure 5: S-shaped PEC cavity. Mono-static RCS versus azimuth angle θ at 6 GHz. Vertical polarisation. RCS convergence using uniform p -enrichments. (a) Uniform $p=4$ and $p=5$. (b) RCS error estimate



(b)



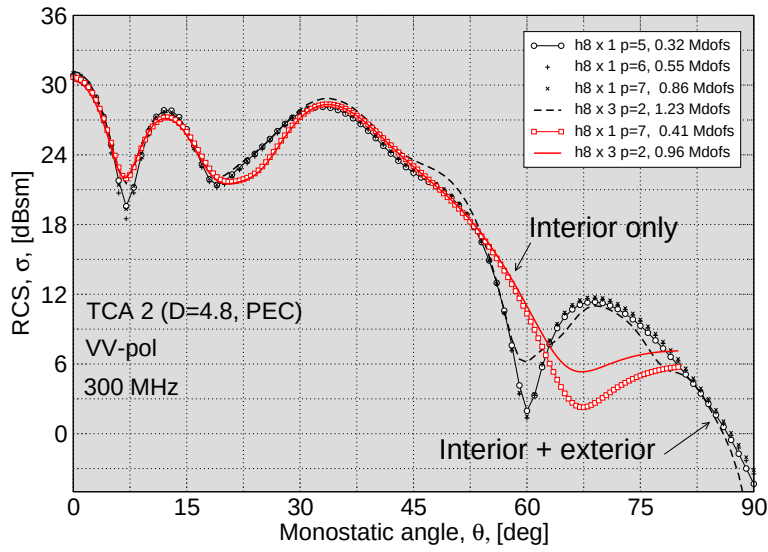
(a)

Figure 6: S-shaped PEC cavity. Mono-static RCS versus azimuth angle θ at 6 GHz. Vertical polarisation. RCS convergence using uniform h -extensions. (a) Uniform $p=4$, and a uniform $p=2$ mesh with a $1 \times h8$ refinement. (b) RCS error estimate

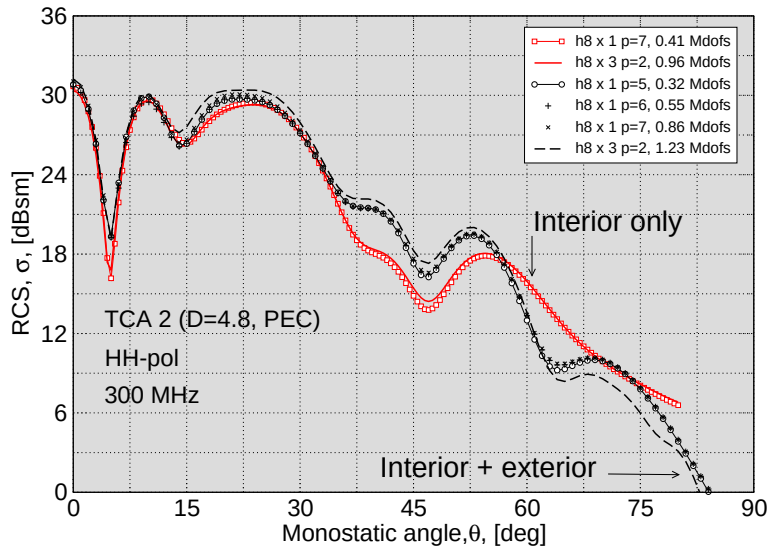
Straight PEC cylinder with a conical termination. We consider the plane wave scattering on a straight metallic (PEC) cylinder of length $L = 5\lambda$ and diameter $D = 4.8\lambda$ with a conical termination. The diameter of the cone is $d = 2\lambda$ and its height is $h = 1\lambda$, see Figure 1. In Figure 7 RCS predictions made with the fully coupled interior plus exterior model are compared to those obtained with the generalised scattering matrix based reduced order, uncoupled model. The fully coupled finite element - infinite element based model includes edge diffraction at the aperture. The aperture rim is sharp in this model. In reality it is not. The RCS for the two models is close up to $\theta = 35$ degrees for horizontal polarisation, and up to $\theta = 45$ degrees for vertical polarisation. This is contrary to the usual rule of thumb that uncoupled models deliver good RCS predictions up to $\theta = 60$ degrees regardless of polarisation as long as the Kirchhoff aperture integration approximation is valid.

Finally, in Figure 8 RCS results obtained with a low order edge based h -type discretization are compared with those obtained with a uniform $p=7$ discretization using large exact geometry elements using the reduced order GSM model, respectively. It is concluded that there is a large potential efficiency gain using higher order finite elements, especially for adaptive hp -FEM. Note that the h -FEM solution has 4.8 Mdof, while the uniform $p=7$ uses only 0.41 Mdof. That is, the

uniform $p=7$ discretization corresponds to one order less finite element degrees of freedom compared to the h -type discretization, giving virtually the same RCS accuracy. Further, note that only 115 guided modes, i.e. generalised degrees of freedom, are sufficient to capture the RCS of this simple channel for close to normal illumination. The number of physical, FE degrees of freedom of the aperture mesh is a couple of orders larger. In other words, the potential for the reduced order model is large with respect to efficiency. The reduction in degrees of freedom also allows solving electrically much larger problems.

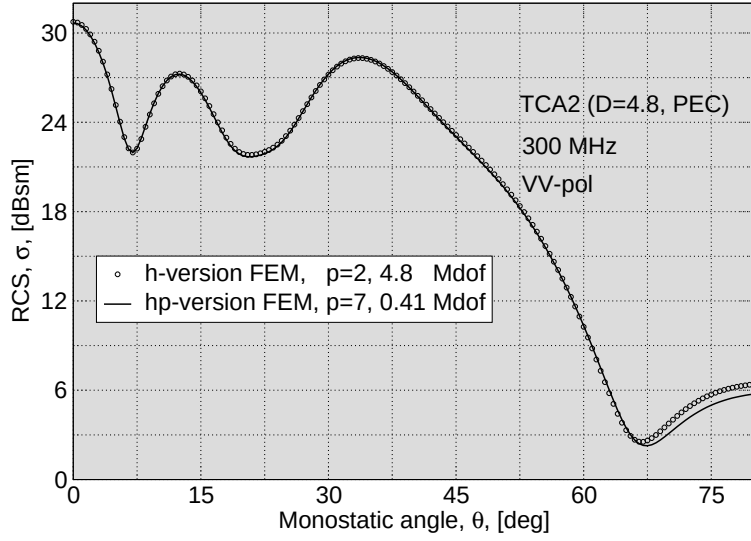


(b)

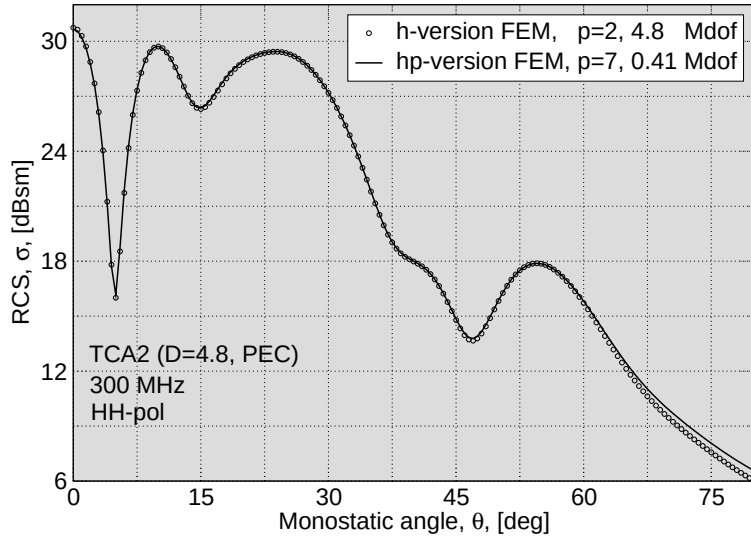


(a)

Figure 7: Straight PEC cylinder with cone. Mono-static RCS versus azimuth angle θ at 300MHz. Full wave, exterior and interior scattering analysis, FE+IE representation, and General Scattering Matrix (GSM) interior only representation using 115 (TE+TM)-wave-guide modes. (a) Horizontal polarization. (b) Vertical polarization



(b)



(a)

Figure 8: Straight PEC cylinder with cone. Mono-static RCS versus azimuth angle θ at 300MHz. General Scattering Matrix (GSM) interior only representation, 115 (TE+TM)-wave-guide modes. (a) Horizontal polarization. (b) Vertical polarization

4. Discussion.

We have presented a full wave theory for the prediction of the so-called cavity RCS with error control applicable to electrically huge semi-opened channels. Arguments and literature references are provided supporting the choice of the *hp*-adaptive FEM method which enables choosing the most efficient discretization assisted by error control. Simple convergence tests with estimates of the discretization error in the RCS for simple uniform mesh refinements and uniform polynomial enrichments, are presented and show clearly in favour of *p*-enrichments. That is, if we only look at the size of the resulting equation system in terms of degrees of freedom. Taking the sparsity of the equation system and the solution time into consideration complicates the picture, and may change it to some extent. Further investigation is needed to shed more light on this issue.

In any case, the prediction of cavity-RCS does not scale with growing electrical channel dimensions. Using Gaussian elimination based equation solution and keeping the reliability requirement on the RCS prediction we indicated that there is a hard bottleneck using one-level domain decomposition based state of the art open-source solvers. As size, taking sparsity into consideration, of the top-level problem grows it will eventually not fit into core, or its solution time

will be unacceptable, if one tries to solve it in situ with a single process. We presented a remedy to this problem. We showed that the size the interface problem can be diminished several order of magnitudes using a generalised scattering matrix based technique. The described bottleneck can be alleviated by executing the interface problem solution separately. Here we reduced the size by using generalised guided wave degrees of freedom. We exploited the fact that the geometry of the channel cross-section provides a natural cut-off frequency based truncation of the two-dimensional series representation of the electromagnetic fields at common sub-domain interfaces. We showed that the RCS can be computed virtually without loss of accuracy using a guided wave representation. We indicated that further gain in efficiency is possible, essentially without loss of accuracy, if we can use the Kirchhoff approximation. We showed that this approximation limits the valid angles of illumination in a problem dependent fashion. The main extra cost for our reduced order modelling technique is the solution of a two-dimensional vector-valued eigen-problem at common interfaces between domains, and one in addition on the aperture of the channel. We presented results based on TE- and TM-travelling waves that are valid for materially homogeneous channel cross-sections. Further investigation is needed regarding guided attenuated waves in materially non-homogeneous cross-sections with dissipative properties.

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